

TABLE OF CONTENTS

M-001 – PIPING

	Page No.
1 SCOPE OF WORK	M 001-4
2 CODES AND STANDARDS	M 001-4
3 DESIGN AND OPERATING CONDITIONS	M 001-5
4 DETAILED REQUIREMENTS	M 001-7
4.1 General	M 001-7
4.2 Piping System Design	M 001-8
4.2.1 Noise abatement measures	M 001-12
4.2.2 Underground pipework	M 001-12
4.3 Miscellaneous Piping	M 001-12
4.3.1 Type of Joint	M 001-13
4.3.2 Flange Joints	M 001-13
4.3.3 Materials	M 001-14
4.4 Corrosion protection and painting	M 001-15
4.4.1 General	M 001-15
4.4.2 Materials and workmanship	M 001-16
4.4.3 Surface preparation	M 001-17
4.4.4 Preparation of coating materials	M 001-19
4.4.5 Application	M 001-20
4.4.6 Protective coatings and paint systems	M 001-21
4.4.7 Galvanizing	M 001-21
4.4.8 Sprayed metal coatings	M 001-22
4.4.9 Weathering steels	M 001-23
4.4.10 Pipelines	M 001-23
4.5 Accessories	M 001-26
4.5.1 Expansion Joints	M 001-26
4.5.2 Traps	M 001-27
4.5.3 General Strainers	M 001-27
4.5.4 Temporary Strainers	M 001-27
4.5.5 Automatic Backwashing Strainers	M 001-28
4.5.6 Flexible Hoses	M 001-29
4.5.7 Restriction Orifice Plates	M 001-29
4.5.8 Sight Flow Indicators	M 001-29
4.5.9 Miscellaneous Piping	M 001-30
4.6 Miscellaneous Accessories	M 001-30

4.7	Identification of Components	M 001-30
4.8	Fabrication	M 001-31
4.8.1	General Fabrication	M 001-31
4.8.2	Shop Fabrication.....	M 001-33
4.8.3	Welding	M 001-34
4.8.3.1	General	M 001-34
4.8.3.2	General fabrication	M 001-34
4.8.3.3	Weld procedure documents.....	M 001-35
4.8.3.4	Weld procedure qualification tests	M 001-36
4.8.3.5	Welders qualification tests	M 001-36
4.8.3.6	Storage of welding consumables	M 001-37
4.8.3.7	Welding equipment	M 001-37
4.8.3.8	Weld examination.....	M 001-37
4.8.3.9	Visual weld inspection	M 001-38
4.8.3.10	Internal examination.....	M 001-38
4.8.3.11	Radiography	M 001-38
4.8.3.12	Ultrasonic examination	M 001-38
4.8.3.13	Magnetic crack detection	M 001-39
4.8.3.14	Dye penetrant tests	M 001-39
4.8.3.15	Quality requirements for welds	M 001-39
4.8.3.16	Mandatory inspections	M 001-39
4.8.3.17	Weld repairs.....	M 001-40
4.8.4	Erection of Piping.....	M 001-40
4.8.5	Flanged Joints	M 001-40
4.8.5.1	Contact Face Finish	M 001-40
4.8.5.2	Gaskets	M 001-41
4.8.5.3	Studs	M 001-41
4.8.5.4	Thread Compound	M 001-41
4.8.5.5	Installation of Studs for High-Pressure Flanged Joints.....	M 001-42
4.8.6	Pipe Supports and Hangers	M 001-42
4.9	Piping Line Number	M 001-42
4.10	Piping Line list.....	M 001-43
4.11	Cleaning, Protection	M 001-43
4.11.1	Cleaning	M 001-43
4.11.2	Protection	M 001-46
4.11.2.1	Materials shipped to the Site.....	M 001-47
4.11.2.2	Materials at the Project Site	M 001-47
5	PERFORMANCE TESTING AND GUARANTEES	M 001-48
6	QUALITY ASSURANCE/QUALITY CONTROL (Q.A/QC).....	M 001-48

EXHIBIT 1	UNIVERSAL DATA SHEET FOR MISCELLANEOUS PIPING SPECIALTIES	M 001-49
EXHIBIT 2	FLEXIBLE HOSES	M 001-50
EXHIBIT 3	EXPANSION JOINTS	M 001-51
EXHIBIT 4	RESTRICTION ORIFICES	M 001-52
EXHIBIT 5	STRAINER	M 001-53
EXHIBIT 6	SIGHT FLOW INDICATORS	M 001-54
EXHIBIT 7	DATA SHEET FOR STEAM TRAPS	M 001-55

M 001 PIPING

1 SCOPE OF WORK

The Bidder shall furnish, the following in accordance with Part 4, Subsection 1.2 “ Summary of Work”, and but not limited to this specification.

During the contract engineering, and not later than 3 months from award of Contract, the Bidder shall supply to the Engineer schedules of the pipework provided under this Contract. These schedules shall state, for each pipework system or parts of a pipework system, the design and operating pressures and temperatures, the fluid transmitted, the piping and valve materials, the types of valves, any corrosion allowances, the pipework design code, insulation proposals, pipe supports and any other data relevant to the mechanical design of the pipework system or part of a pipework system.

All necessary main and miscellaneous piping, vents, drains and piping accessories and appurtenances shall be furnished, fabricated, delivered and installed by the Bidder to make a complete, safe and operable piping system.

Bidder shall also provide suitable unions or flanged couplings for all traps, float cages, filters, strainers and other similar appurtenances for ease of maintenance and removal.

All components shall be of manufacture and type indicated in the Specifications and shall be of material suitable for the design conditions and intended purposes, and shall be acceptable to the Owner.

Instrument Connections, In line devices (e.g., thermowells, test wells, flow nozzles, control valves, etc.) shall be in accordance with requirements of Control and Instrumentation Specification.

2 CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5, “Codes and Standards”. The codes and standards of the following organizations extracted from Part 4, Subsection 2.5 “Codes and Standards” or other equivalent international standards are specifically applicable to the design, manufacture and testing of the equipment included in this specification.

The design, materials, fabrication, construction, testing and certification of all pipes, valves, piping components and supports shall comply with the latest revision of ANSI / ASME B31 .1 - Power Piping Code, including all addenda (Referred to herein after as the Code) Unless otherwise specifically noted, and also shall comply to all Indonesian Codes and Regulations having jurisdiction.

The Owner shall be notified in writing for resolution of any discrepancies between the requirements of this specification, or portions of other

specifications within Part 4 which relate to piping systems, and the specified design Codes and Standards. The Bidder shall clearly describe the discrepancy and the proposed resolution to the discrepancy. The Bidder shall not proceed prior to written instructions from the Owner.

All ASME Code stamp for pipe including Boiler external piping, valves and other components furnished by the Bidder shall be marked and stamped in accordance with the applicable requirements of the appropriate ASME Code, when required by the Code.

3 DESIGN AND OPERATING CONDITIONS

All piping shall be routed to provide a neat and economical layout and requiring the minimum number of fittings. Piping shall be arranged so that full access is provided for the operation and maintenance of equipment, and that removal or replacement of equipment can be achieved with the minimum dismantling of piping. Overhead piping, inclusive of fittings, insulation and support steelwork, shall have a minimum vertical clearance of 2.3 meters above access platform and walkways and 6 meters above roadways.

In this and other parts of this Specification the term "high pressure" refers to steam pipework where the design pressure exceeds 17 bar and to other systems where these values exceed 24 bar.

The following standard pipe sizes shall be used for the different systems:

Diameter in DN: 25, 50, 80, 100, 150, 200, 250, 300, 350, 400 etc.

Pipes with a diameter size smaller than DN 25 may be used only for control air, control oil, chemical dosing, sealing water, instrument connections, cooling water connections to equipment, sampling pipes and input pipes.

All drain and vent valves shall be located in easily accessible positions. At least one valve shall be Suitable for regulation at maximum service pressure. Drains pipework shall have a continuous fall to the point of discharge. All drains and vents shall discharge to safe locations. Drains and vent pipework shall be neatly run and shall not interfere with maintenance or operating access requirements.

The nominal diameter of drains and vent pipework shall be suitably sized according to requirement.

As far as it is possible all process pipes shall be above ground on pipe supports. Wherever the pipes have to cross roads or other facility, this shall be achieved by proper trenches. No buried pipes are acceptable for fuel oil.

The Bidder shall establish the design ratings for piping systems based on maximum coincident pressure and temperature conditions within the system.

The Bidder shall complete sets of all isometric drawings in an agreed format A3 or A2 for each system, properly classified and will keep it continuously updated with the latest revisions of the isometrics.

The layout of the piping shall be subject to approval by the Owner/Engineer. These drawings shall indicate the position of all supports, drainage points, fittings and all closing pipes. However, all piping systems shall be arranged to allow for adequate slopes in the direction of flow. At the point of drainage, drain pockets of sufficient size and approved construction shall be fitted. Drain piping shall be separately run to the blowdown tank or drain tank as may be appropriate.

Design conditions in all pump discharge piping shall consider operation at shut off conditions, regardless of any recirculation systems, and shall include suitable margins to account for pump manufacturing tolerances and safety valve tolerances (where applicable). For systems at elevated temperature, the Bidder shall consider operation with cold fluid during start-up and flushing activities, as appropriate.

Design conditions for main steam piping including valves shall be in accordance with ANSI B31.1, "Power Piping Code" requirements.

Design conditions downstream of desuperheating devices shall consider the length of piping required for through mixing of high temperature steam and the desuperheating (cooling) medium.

Low pressure piping downstream of all pressure regulating devices shall be protected by safety or relief valves unless the downstream piping is designed for the same conditions as may exist upstream of the pressure regulator. Safety or relief valve capacity shall consider the pressure regulator fully open and the low pressure system blocked in (where such block valves exist).

Fluid velocity in all systems shall be within the maximum velocity criteria shown in velocity table, below :

<u>Fluid</u>	<u>Velocity m/sec</u>
Steam : Saturated	20 - 30
Superheated	30 - 45
Superheated (300 mm and larger in size)	50 - 70
Exhaust to atmosphere	25 - 50
Water : Pump inlet	2 - 2.5
Pump Outlet less than 10 bar	2.5 - 3
Pump Outlet 10 bar or more	3 - 3.5
Drain	1 - 2
General Service	1 - 3
Air : Low Pressure Air (less than 10 bar)	10 - 20
High Pressure Air (10 bar or more)	10 - 15
Compressor inlet	10 - 20
Oil : Fuel oil and lubrication oil	0.5 - 2
N. Gas : Natural Gas to G/T	20 - 40

Piping which contains corrosive fluids, such as concentrated acid or caustic, shall not be routed over electrical equipment or cable trays. Joints shall not be located where such piping is routed over walkways.

4 DETAILED REQUIREMENTS

4.1 General

Piping shall be designed, fabricated, examined and tested in accordance with the following:

- The governing code(s) specified above.
- The Owner's drawings forming a part hereof.
- The Piping Line List to be submitted by the Bidder per Clause 4.10
- All piping over DN 50 shall be shop fabricated and smaller piping shall be shop fabricated as far as possible to reduce the number of field joints to a minimum. Design drawings shall be submitted to the Owner for review prior to fabrication.
- Other requirements set forth in this specification.

For the design of safety valves installed downstream of reducing stations, high pressure bypass valves or equivalent control valves, the maximum throughput of the fully open reducing or bypass valve including injection water quantity is to be taken as the basis for calculation. All cross sections and lines for safety

devices that protect against excess pressure (safety valves, rupture discs and similar items) must be designed to ensure the necessary blow-off rate and fault free functioning.

The design pressure is equal to the response pressure of the devices protecting the piping. In the case of pump discharge lines a pressure corresponding to 1.1 times of the shut-off head of the pump at the ambient temperature is to be taken as the minimum design pressure. The main cooling water systems must be designed according to the water hammer calculation.

The design temperature is the highest possible fluid temperature occurring in the length of line concerned. Possible tolerances of the temperature control system and any temperature allowances provided by the requirements of the standards shall be considered.

For live steam lines the design pressure of the lines shall be equal to the response pressure of the protection device protecting the piping. The pressure of the superheater safety valve shall be established based on the margins of set points of bypass valves over turbine inlet pressure and the pressure drop in the main steam line.

In addition to the required wall thickness in accordance with calculations, a corrosion allowance of 1 mm shall be added for unprotected carbon steel, 1 mm for alloy steel, and 0 mm by stainless steel.

All elbow fittings shall have a wall thickness equal to or greater than the wall thickness of the matching pipe. In no case shall the elbow wall thickness be greater than 1.25 times the matching pipe minimum wall thickness.

All instrument connection shall be as per following:

- (a) All instrument connections shall be provided with a root valve of minimum DN15 located close to and in accordance with the design requirements of the piping main where the instrument is located. The root valve shall not be a substitute for instrument isolation valves which shall be provided near each instrument.
- (b) Tube fittings and joints shall be by a single manufacturer for all plant tubing, including tubing supplied within all equipment packages. The selected tube fitting and joint system manufacturer shall be approved by the Owner.
- (c) Double isolation valves shall be provided for all steam and feed water vents and drains, and for all instrument connections, in piping with rating above PN 40.

4.2 Piping System Design

- Design and construction of all parts of the pipelines and accessories should correspond to the present state of the art and shall be based on the latest standards.

- The pipe work and its accessories shall be designed and arranged so that all parts subject to operation and maintenance can be operated, inspected, maintained and replaced without difficulty and with a minimum of effort. All important parts must be accessible.
- Provisions to allow for isolation and for access must be foreseen on all parts subject to acceptance resets by the local authorities.
- The nominal bore of gate valves shall be the same as the nominal diameter of the pipeline in which they are installed.
- None of the forces and moments transmitted by the pipes to connected machines, apparatus and other components must exceed the maximum permissible values, given by the manufacturers of these items. Attachments to turbine foundations shall be carried out only as agreed with the turbine manufacturer.
- All steam traps shall be provided with a bypass and lines which open to a funnel.
- As far as expansion joints and other parts of pipe work are concerned it shall be borne in mind that differential settlement can occur. The reaction forces and moments of the piping system to be withstood by fixed points, walls, foundations and other civil structures shall be reduced to the utmost minimum by suitable means (e.g. expansion joints shall be provided where required).
- Since the steam pipes may be filled with water for a short time (e.g. when flushing or pressure testing), supports shall be dimensioned accordingly. The load information to be given by the Bidder to the civil designers shall take account of this requirement.
- The pipe support structures shall be designed to minimize heat transfer.
- The installed pipe work with its supports and other components shall not obstruct gangways (min. 1000 mm wide), maintenance, escape routes etc. Overhead piping shall have a minimum vertical clearance of 2.3 meters and 6 meters above roadways.
- Pipe spools for HP steam, feed water piping and all steel pipes shall be cleaned internally prior to delivery by shot blasting at the workshop with iron particles to SA2 ½ or by acid cleaning, and shall be properly protected against corrosion.
- Pipe ends and branch connections of underground piping shall be sealed temporarily during installation if the connecting pipe is not immediately installed.
- The Bidder shall submit a detailed description of proposed steam blowing and other cleaning procedures for all pipelines, and no part of this work shall be started until these procedures have been approved by the Owner. Temporary silencers shall be provided for this.
- The Owner reserves the right to require the Bidder to modify any of his cleaning procedures if found necessary to obtain acceptable results. The

Bidder shall furnish, install and dismantle all temporary pipes, hangers, anchors, etc. required for cleaning all piping systems.

- To the extent that hot fluids can accumulate in pipe sections isolated for maintenance purposes (including control valves with injection water), drains with hand-operated shut-off valves are to be provided for the safety of the personnel (block and bleed systems) . Furthermore the Bidder shall safeguard the piping systems against over-pressurization caused by thermal expansion of blocked-in fluids by adequate means.
- All welding shall be carried out according to relevant standards. For quality reasons as many welds as possible are to be carried out in the workshop.
- Welding ends of all piping must be carefully prepared before welding. The type of butt welding ends of valves, control devices, orifices, etc. shall be specified by the Bidder and must be given to the manufacturers of these components in due time prior to their start of work if necessary post-weld heating treatment. If there are differences in the wall thickness and/or different materials of piping and valves with butt-welding ends, the necessary transition pieces must be provided by the manufacturer of the valves.
- Socket welds are not permitted for lines above DN 50, for corrosive media lines or for lubricating oil lines.
- After completion of the weld joint, the welder must mark with indelible crayon his identity number and the last two digits of the year in which the work was completed on the pipes. The Bidder shall keep a record of welders.
- A Stress analysis must be performed for all piping systems DN > 80 with an operating temperature > 120°C. The piping stress and flexibility analysis shall be based on the relevant standards. Recalculation must be taken into consideration as built condition, actual weight and dimensions.
- Pipe materials, bends and fittings shall be tested in accordance with the applicable material standards.
- HP steam and feedwater pipes > DN 50 shall be 100% radiographed. Remaining components shall be tested to 10% per weld as a minimum.

Piping Line Identification shall be per Part 4, Subsection 2.8, Plant Identification. Unless otherwise indicated, each piping line on the piping and instrument diagrams (P&ID's) and/or the piping physical drawings shall bear an encircled identification figure (line number), except certain small low-pressure/low-temperature lines such as instrument air branch lines and draft gauge and VAC air sensing lines collectively identified and specified by notes on the drawings. Changes in material, pressure class, size, pipe wall thickness, code jurisdiction, etc., in a line shall be shown on design drawings at the actual point of change.

The line numbers shall be used for identification of the lines in all documentation.

All standard components of the piping systems shall conform to the applicable specifications and standards listed in the governing code(s).

Piping and appurtenances which are indicated on the design drawings but are not specifically mentioned in the Specifications, shall be of such pressure standards specified for other piping and appurtenances for similar pressure, temperature and other conditions existing and shall be acceptable to the Owner.

Unless otherwise indicated, all vents and drains from piping and equipment upstream of the last valve) shall be of the same material and pressure standard as specified for the corresponding system to which the vents and drains are connected.

The last valve in a vent or drain line is the valve that may be subject to the full system pressure on its inlet side and atmospheric (or near atmospheric) on the outlet side.

The type, class, grade, or quality of material proposed by the Bidder, will be reviewed by the Owner for approval.

The steam and drain piping shall conform to the following

- (a) Materials: Materials for pipe, fittings, valves, flanges, etc., shall be as specified in Clause 4.3.
- (b) Pipe: Seamless piping material unless specified otherwise.
- (c) Fittings: Welding fittings shall be of equal or greater wall thickness than the connecting pipe.
- (d) Bolting for Flanged Piping Joints: Bolting for steam service pressure above 17 bar or steam-service temperatures above 230°C shall be alloy steel as approved. All others may be carbon steel.
- (e) Nuts for Alloy Steel Studs: Oil-quenched, hot-forged, medium carbon steel, semi finished, heavy hex nuts, per ASTM Specifications A 194, Grade 1 or 2 for service temperatures up to 400°C. Grade 2H for higher service temperatures. ANSI Class 2B fit.

- (f) Nuts for Carbon Steel Bolts: Grade 1 , cold-punched, carbon steel, semi finished, heavy hex nuts, per ASTM Specifications A 194, ANSI Class 2B fit.
- (g) Socket welded connections may be used in sizes larger than 50 mm for special connections at equipment where it may be more suitable for a equipment or piping, they shall be made with heavy steel tubing, preferably specific location than a butt welded joint if approved by the Owner.

4.2.1 Noise abatement measures

- The Bidder shall take all necessary measures to limit noise in accordance with the latest developments in technology. On no account shall the sound level exceed the values mentioned under the relevant subsection .
- If this requirement cannot be met by adequate construction of the pipework and valves concerned, sound absorbing housing or insulation have to be provided.
- Where pipework and valves are installed outdoors, the above mentioned requirement may be fulfilled, if necessary, by providing silencers at the safety valve outlets and at the blowing out lines.

4.2.2 Underground pipework

All underground pipework shall be designed in order to withstand the loading from covering soil as well as from traffic.

Underground piping systems shall be self-anchoring, i.e. no thrust blocks for loadings resulting from test and operation pressure are permitted except in specific areas where partially encased thrust blocks may be provided as reviewed case by case.

The outside insulation of such pipework has to be selected in order to withstand the prevailing highly corrosive conditions. The water conditions are described in the general project requirements.

All applicable steel underground pipe work shall be protected by cathodic protection and factory applied wrapping. All underground pipe routes shall be indicated by above ground markers to the satisfaction of the Owner.

Where guiding tubes or sleeves for penetrations such as through building walls, walls or dikes of tank farm, etc., of steel are practicable, such material has to be supplied and proper sealing and outside protection carried out by the Bidder.

4.3 Miscellaneous Piping

- (a) All miscellaneous low pressure piping not specified elsewhere, in size 80 mm and larger shall consist of seamless steel pipe of not less than standard wall thickness with steel butt welded fitting and with butt welded pipe joints, except where flanged joints may, of necessity be used.

- (b) Pipe flanges, where used, shall be of forged steel slip-on or welding-neck type. Piping in size 50 mm and smaller shall consist of seamless steel pipe according to design requirement
- (c) No cast iron or malleable-iron fittings shall be used.
- (d) Gauge lines shall not be smaller than 15 mm (except a reduction to 8 mm size at the instrument is permitted, if necessary) and shall be constructed of heavy materials. Where these connections are made to any part of the equipment or piping, they shall be made with heavy steel tubing, preferably with socket welded joint and with a steel shut off valve located at the equipment or piping. Substantial bosses shall be provided on equipment piping for all instrument connections.
- (e) All fittings, flanges, facing, bolting etc. shall conform standards for the pressure standard involved in each case.

4.3.1 Type of Joint

All joints at fittings, equipment, etc., shall be welded, insofar as possible. Flanged joints shall be used only where necessary and in all cases only when acceptable to the Owner.

4.3.2 Flange Joints

- (a) Pressure Standard: As required by ANSI B 16.5 and based on pressure and temperature existing at maximum turbine rating.
- (b) Flanged Facings:
 - For ANSI Class 150 and 3001.5 mm raised face.
 - For ANSI Class 400 through 2500 ...6 mm raised face or manufacturer standard ANSI flange facing suitable for the pressure / temperature condition
- (c) Gaskets : Suitable gaskets shall be provided at all flanged joints.

4.3.3 Materials

All material for pipe, fittings, flanges etc. shall be as the following requirement:

Classification of Piping Materials and Design Limits shall be as per following table.

Classification (ASTM)	Max. working pressure Bar	Max. working temperature ° C	Use for
A53	10	350	Water, oil, air
A139	10	350	Water, oil, air
A-333	10	350	Fuel oil, water
A-106 Gr.B,C	No limitation	450	Fuel oil, water, Steam, Natural gas
A-335 Gr.P1	No limitation	500	Water, Steam
A-335 Gr.P1 2	No limitation	550	Water, Steam
A-335 Gr.P11	No limitation	575	Water, Steam
A-335 Gr.P22	No limitation	600	Water, Steam
A691 1 ¼ Cr CL22	No limitation	575	Water, Steam

Fuel oil piping shall be of carbon steel.

The piping for the closed cooling water or fire protection water shall be made of carbon steel.

The drain piping of boiler blow down tank shall be of A106 Gr. B and thickness should not less than Schedule 40.

The chlorine and chloric solution piping shall be of polyvinyl chloride or rubber lined carbon steel piping.

The sampling piping for water and steam shall be stainless steel.

Instrument air main piping shall be of galvanized carbon steel or cooper. The shop joints of piping shall be of welded type before galvanizing. The field joints shall be either flanged or thread type joints. No welding shall be permitted after galvanizing.

The water piping for drinking water shall be made of galvanized carbon steel.

The turbine oil piping shall be made of steel pipe of SA 333 Gr. 1. and Schedule 40, or stainless steel. The connection shall be of welded type as far as possible.

The sea water piping shall be glass-fiber reinforced pipe (GRP/FRP). Alternatively Carbon Steel piping can also be accepted with proper protection to against the corrosion caused by sea water.

Piping materials under special conditions shall be as follow:

- | | | |
|-------------------|---|--|
| Sodium hydroxide | - | Stainless steel |
| Hydrochloric acid | - | Rubber lined carbon steel pipe |
| Other chemicals | - | Rubber lined carbon steel or polyvinyl chloride pipe |

For the piping of drains vents, instruments, etc. the same materials as the main piping up to the shut-off valve on the branch piping. For HP steam system, this clause applies to the second shut-off valve.

All materials not specified shall be in accordance with the applicable codes. All special pipe, fittings, flanges etc. not covered by this specification or the governing code shall be subject to review and acceptance by the Owner. All stainless steel piping materials, which are welded, shall be type "L", e.g. 304 L, or 316 L.

4.4 Corrosion protection and painting

4.4.1 General

The intent of this specification is to define surface preparation, protective coatings and paint systems to be applied on vessels, equipment, piping and structures.

The specification shall be used as a basis for the preparation of painting schedules, engineering drawings and requisitions of materials and labor.

This specification covers the general requirements for painting work of the project. This specification also outlines the type of paint, surface preparation, touch up for prime coat(s), number of prime and finish coats, and dry film thicknesses for any item requiring painting on the project.

The Bidder shall furnish all labor, materials and equipment including paints and thinners of the best quality. The proposed methods shall be in accordance with the best engineering practice and to the approval of the Owner or his Representatives.

The general guides for the work shall be :

- ISO 12944, ISO 8501
- This Specification
- British Standard BS 5493
- Swedish Standard 055900
- German Standards DIN 55 928 and DIN 4115
- The paint manufacturer's product data sheets and instructions for paint application
- European Committee of Corrosion Protection Standard
- All regulations concerning safety of work with paints and coatings

4.4.2 Materials and workmanship

The materials and equipment used, the methods of application and the quality of work shall at all times be subject to the inspection and approval of the Owner or his Representatives. Primary coats of paint should not be applied until the surface preparatory work has been done according to the applicable codes and standards.

The Bidder and/ or paint supplier shall assist the Owner in making any tests or inspections when, in the opinion of the Owner such inspection is necessary or specified.

No consecutive coats of paint, except in the case of white, shall be of the same shade.

All paint film thicknesses quoted in the attached table "Paint System" refer to the dry film condition. Where the surfaces to be painted are subject to heat or attack chemicals, oil, acid, fumes, etc. special paint having the appropriate corrosion resistant qualities shall be used.

Defects which affect the corrosion protection and which are caused whether by inadequate workmanship of application or by the coating material supplied by the sub-Bidder, must be rectified by the Bidder at his cost.

The Bidder shall consider a 10 year design guide for the corrosion protection using high quality coatings with synthetic resins with the need for only minor remedial work in the guarantee period by the Client. A design guide of 4 years must be given for hot-applied bituminous coatings, high-build bituminous

coatings (film thickness 300 microns) or coating systems with materials of non-bituminous basis but with the same corrosion resistance.

A guarantee period of 3 years shall apply for the high-build bituminous coatings (film thickness 300 microns) or coating with materials of non-bituminous basis but with the same corrosion resistance.

The plant shall be inspected at the end of warranty period and areas of rust shall be repainted by the Bidder. If the rust is extensive, then all the relevant system area shall be repainted as mutually agreed between the Bidder and the Owner

Painting shall be done only when the surface temperature is above 5°C. Surface temperature must be at least 3°C above dewpoint to ensure that condensation does not occur on the surface. Relative humidity must be checked to prevent condensation on the surface.

Reasonable caution shall be exercised to avoid the painting of outdoor parts etc. Immediately prior to rain.

In hot weather, precautions shall be taken to ensure that the specified dry film thickness of priming or finish coats is obtained.

Any prime coat exposed to excess humidity, rain etc. Before drying shall be permitted to dry and the damaged area of primer removed and the surface prepared and primed again.

Sheltered or unventilated horizontal surfaces on which dew may collect require more protection and to achieve this an additional top coat of paint shall be applied.

The temperature quoted as “normal” in the “Paint System Tables” refers to the climatic conditions encountered in the area of the projects.

The Paint System Schedules attached in the Annexes are the minimum requirements.

4.4.3 Surface preparation

The life of a coating system depends primarily on surface preparation. Surface preparation, therefore should remove sufficient foreign matter to allow the type of prime paint used to wet the surface thoroughly and develop adequate adhesion.

In preparing any surface to be coated, all loose paint, dirt, grease, rust, scale, weld slag or spatter or any other extraneous material shall be removed and defects repaired, so as to obtain a clean, dry, even surface to receive the priming or finishing coat(s) as called for in the painting schedules. Sharp edges should be rounded, especially when tank linings have to be applied.

All machined surfaces including flange faces shall be suitably covered to prevent damage during surface preparation.

All surfaces shall be blast cleaned whenever possible.

Bare steel surfaces shall be prepared by one of the methods described below in order to preference and in accordance with Swedish Standard SIS 05 59 00 or Steel Structures Painting Council (SSPC) or ISO 8501.

White metal blast cleaning

Blast cleaning to bare metal. Mill scale, rust and foreign matter must be removed completely. Subsequently, the surface is cleaned. It must then have a uniform metallic color and correspond in appearance to the prints designated SA3 (or SP5 - SSPC).

Near white metal blast cleaning

Very thorough blast cleaning. Mill scale, rust and foreign matter shall be removed to the extent that the only traces remaining are slight imperfections in the form of spots and stripes. It must then correspond in appearance to the prints designated SA 2 1/2 (or SP10 - SPCC).

Very through mechanical scrapping and wire brushing

Very thorough scrapping and wire brushing - machine brushing -grinding - etc. surface preparation as for St 2 but much more thoroughly. After the removal of dust, the surface have a pronounced metallic cheen and correspond to the prints designated St 3.

Mechanical cleaning should only be used when procedures (a) and (b) are not practicable.

Thorough scrapping and wire brushing

Thorough scrapping and wire brushing - machine brushing-grinding-etc. The treatment shall remove loose mill scale, rust and foreign matter. It should then have a faint metallic sheen. The appearance must correspond to the prints designated St 2.

The abrasives used for the blast - cleaning shall be graded flint, grit, short or silica and shall be such that they will produce an average keying profile on the blast - cleaned surface of not less than 40 microns.

An air pressure of 7 bar g at the nozzle shall be used.

After blast cleaning all accumulated grit, sand, dust etc. must be removed leaving the surface clean, dry and free of mill scale, rust, grease and other foreign matter.

In the event of rusting after completion of the surface preparation, the surface must be cleaned again in the manner specified.

Oil, grease, soil, cement, salts, acids or other corrosive chemicals shall be cleaned from steel surfaces, by the use of solvents, emulsions or cleaning compounds. The final wiping shall be with clean solvent and clean rags or brushes. There shall be no detrimental residue left on the surface.

Primed areas which suffer damage must be shot blasted on site before touching up.

Protective coating must be applied as quickly as possible after the completion of surface preparation no matter what cleaning method has been used.

No blast-cleaned surfaces shall be allowed to remain uncoated overnight.

Steel work protected by shop primer after arrival on site must be cleaned of salt, sand, oil etc. Before the first coat of paint is applied on site. Shop primer damaged during transport must be rectified by blast cleaning and coating before application of the site coats.

4.4.4 Preparation of coating materials

All containers shall remain unopened until required for use. Primers and paints which have livered, gel led or otherwise deteriorated shall not be used.

The oldest primer or paint of each kind shall be used first.

All ingredients in any container shall be thoroughly mixed before use, and shall be agitated frequently during application to keep the primer in suspension.

Primer or paint mixed in the original container shall not be transferred until all settled pigment is incorporated into the body of liquid.

Mixing in open containers shall be done in a well-ventilated area.

Primer or paint shall be mixed in a manner ensuring the breakdown of all lumps, complete dispersion of pigment and uniform composition.

Two-component primers shall be mixed in accordance with the manufacturer's instructions.

Thinners shall not be added to primers or paints unless necessary for proper application according to the manufacturer's instructions.

The type of thinners used must comply with the manufacturer's instructions.

When use of thinners is permitted, it must be added to the primer or paint during mixing.

4.4.5 Application

All prime coatings shall be applied by brush or airless spray or a combination of these methods, as approved by the coating manufacturer.

Spray guns shall not be used outside in windy weather or near surfaces of a contrasting color unless the latter are properly protected from the spray.

All cold-spray painting shall be done using standard equipment in accordance with accepted industry standards and methods.

Paint erroneously applied to items that are not to be painted shall, if the Engineer so demands, be removed at the Bidder's expense, leaving the surface clean, unstained and undamaged.

Dry film thickness (dft)

To the maximum extent practical, the coats shall be applied as a continuous film of uniform thickness and free of pores. Overspray, skips, runs, sags and drips should be kept to a minimum. The different coats shall not be of the same color.

Each coat of paint shall be allowed to harden before the next is applied. For epoxy paint the hardening time before the next coat shall normally be 12-14 hours. Supplier's recommendations regarding hardening time of epoxy paints must be followed.

Particular attention must be paid to full film thickness at edges.

The minimum total dry film thickness of the paint systems shall be as recommended in the following tables. The dft is given in microns (millionths of a meter).

4.4.6 Protective coatings and paint systems

A color scheme shall be established by the Bidder in co-operation with the Engineer (based on RAL standard).

The type and number of protective coats for any item requiring painting on the project shall be in accordance with the attached tables "Paint Systems" and the Owner's requirements.

Generally, all parts shall receive the specified prime coat(s) at the supplier's works to avoid corrosion during transport to the site and storage at the site. Parts which cannot be damaged during transport shall receive the full number of coats.

A UV protection final coating using a polyurethane based paint is preferred.

4.4.7 Galvanizing

Galvanizing work shall conform in all respects to BS 729, BS 3083 and BSCP 2008 and shall be performed by the hot dip process, unless otherwise specified.

Cable trays, tray covers and associated equipment shall be hot dip galvanized.

It is essential that details of steel members and assemblies which are to be hot-dip galvanized should be designed to suit the requirements of the process. They should be in accordance with BS 4479 and DIN 55928.

Vent holes and drain holes shall be provided to avoid high internal pressures and air locks during immersion, which may cause explosions, and to ensure that molten zinc is not retained in pockets during withdrawal.

Careful cleaning of welds is necessary before welded assemblies are dipped. The welds and the surrounding metal shall be cleaned separately, preferably by blast-cleaning, because the usual preliminary pickling cannot be relied on to remove the welding slag.

All defects of the steel surface including cracks, surface laminations, laps and folds shall be removed in accordance with BS 4360. All drilling, cutting, welding, forming and final fabrication of unit members and assemblies shall be completed, where feasible, before the structures are galvanized. The surface of the steel work to be galvanized shall be free from paint, oil, grease and similar contaminants.

The minimum average coating weight shall be as specified in Table 1 of BS 729. Structural steel items shall be first grit-blasted to BS 4232, second quality, (Sa 2 1/2) or pickled in a bath, and the minimum average coating

weight on steel sections 5 mm thick and over shall be 900 g/m², on steel sections 2 - 5 mm thick 600 g/m².

On removal from the galvanizing bath, the resultant coating shall be smooth, continuous, free from gross surface imperfections such as bare spots, lumps, blisters and inclusions of flux, ash or dross.

Galvanized contact surfaces to be joined by high-tensile friction-grip bolts shall be roughened before assembly so that the required slip factor (defined in BS 3294, Part 1 and BS 4604, Part 1) is achieved. Care shall be taken to ensure that the roughening is confined to the area of the mating faces.

Bolts, nuts and washers, including general grade high-tensile friction grip bolts (referred to in BS 3139 and BS 4395, Part 1) shall be hot-dip galvanized and subsequently centrifuged (according to BS 729). Nuts shall be tapped up to 0.4 mm oversize after galvanizing and the threads oiled to permit the nuts to be finger-turned on the bolt for the full depth of the nut. No lubricant, applied to the projecting threads of a galvanized high-tensile friction-grip bolt after the bolt has been inserted through the steel work, must be allowed to come into contact with the mating faces of the steel work.

Protected slings must be used for off-loading and erection. Galvanized work which is to be stored at the works or on site shall be stacked so as to provide adequate ventilation of all surfaces to avoid wet storage staining (white rust).

Small areas of the galvanized coating damaged in any way shall be restored by:

- cleaning the area of any weld slag and thorough wire brushing to give a clean surface.
- the application of two coats of zinc-rich paint, or the application of a low melting point zinc alloy repair rod or powder to the damaged area, which is heated to 300°C.

After fixing, bolt heads, washers and nuts shall receive two coats of zinc-rich paint. Connections between galvanized surfaces and copper, copper alloy or aluminium surfaces shall be protected by suitable tape wrappings to the approval of the Owner/Engineer.

4.4.8 Sprayed metal coatings

Corrosion protection can also be achieved by spraying aluminium, zinc, tin, copper, lead or other suitable metals on the surfaces of structures. Composition of coating metal s, methods of surface preparation and application of coatings, requirements for thickness and adhesion and subsequent treatment shall conform to BS 2569.

The requirements for sprayed, unfused coatings of metals and metallic compounds applied by combustion gas flame, plasma arc, detonation and

similar processes, and the preparation of components, spraying techniques, sealing, finishing and inspection are those of BS 4761.

4.4.9 Weathering steels

The requirements of Standards ASTM-A 242, ASTM-A 588 and DIN 17 100 must be observed in the use of structural steel which is corrosion-inhibited, so-called weathering steels.

4.4.10 Pipelines

All pipelines shall be marked according to the medium transported in agreement with Color Code approved by the Engineer. The marking shall be executed by painted color rings in suitable size and number.

Coating Systems								
System-No.	Surface Location	Temp.°C	Surface Preparation	Coating Systems	No. of coats	Generic Type	Dryfilm thickness (DFT) per coat µm	Total DFT µm
1	Structural steel works, piping, vessels, tanks INDOOR	up to 120	SA 2.5	Primer	1	Zinc-Epoxy	80	80
				Finish	1	Epoxy High Solid	80	<u>80</u> 160
2	Structural steel works, piping, vessels, tanks OUTDOOR	up to 120	SA 2.5	Primer	1	Zinc-Epoxy	80	80
				Intermediate	1 – 2	Epoxy High Solid	160	160
				Finish	1	2-Comp.Polyurethane	50	<u>50</u> 290
3	Piping, tanks, etc. INDOOR and OUTDOOR, Insulated	up to 120	SA 2.5	Primer	1	Zinc-Epoxy	50	50
4	Pumps, motors, other equipment OUTDOOR	up to 120	SA 2.5	Primer	1	Zinc-Epoxy	80	80
				Intermediate	1	Epoxy High Solid	110	110
				Finish	1	2-Comp.Polyurethane	50	<u>50</u> 240
5	Pumps, motors, other equipment INDOOR	up to 120	SA 2.5	Primer	1	Zinc-Epoxy	80	80
				Finish	2	Epoxy High Solid	50	<u>100</u> 180
6	Piping, reactors OUTDOOR Insulated	> 120	SA 2.5	Primer	1	Zinc Ethysilicate	75	75

Coating Systems								
System-No.	Surface Location	Temp.°C	Surface Preparation	Coating Systems	No. of coats	Generic Type	Dryfilm thickness (DFT) per coat µm	Total DFT µm
7	Stacks OUTDOOR	< 120	SA 2.5	Primer	1	Zinc Ethysilicate	75	75
		< 200		Finish	2	Silicone Acrylic	50	<u>100</u> 175
8	Steel surfaces Uninsulated	200 – 450	SA 2.5	Primer	1	Zinc Ethysilicate	75	75
				Finish	2	Silicone Aluminum	25	<u>50</u> 125
9	Galvanized surfaces	up to 120	Mechanical cleaning from contaminants and zinc salts by means of washing or steam jetting or sweep-blasting with fine sand	When Finish Coat is required, such as sea climate with chloride exposure	1	Epoxy High Solid*	125	125
10	Steel surfaces permanently in contact with water, also seawater splash zone	Medium temp. °C up to 60	SA 2.5	Prime and Finish Coat in One	1	Glassflake reinforced High Solid Epoxy	500	500

m) Additional 1 x Finish Coat 2-Comp. Polyurethane, 50 µm, when exposed to UV and colour retention is required or when exposed to weathering.

4.5 Accessories

4.5.1 Expansion Joints

- (a) Expansion joints shall be of metal or rubber.
- (b) Each expansion joint shall be suitable for the working pressures or vacuum, temperatures, and pipe movements in the line where it is to be installed and shall be of design, type and manufacture acceptable to the Owner. Local manufacturer with international quality standard for easy maintenance are preferable.
- (c) Description of the exact type of joint for each individual location as well as the details of joint reinforcement for pressure or vacuum service shall be reviewed by the Owner.
- (d) Flanges shall have dimension and drilling in conformance with the pressure standard for the line in which the joint is to be installed. For non standard sizes, dimensions and drilling shall be as detailed on the drawings and/or as approved by the Owner.
- (e) Restraining rods, where required, shall be designed to resist 150% of the joint pressure area force. Restraining rods or tie bars shall be provided with lockouts in addition to normal nuts so that a total of six (6) nuts per restraining rod are furnished.
- (f) Metal expansion joints shall be of corrugated stainless steel, type 304 for temperatures up to 425°C and Type 321 for temperatures of 425°C and higher. Metal expansion joints shall be provided with internal sleeves of same material as the corrugated member and shall comply with the following requirements.
- (g) Metal joints shall be provided with suitable reinforcing rings, where required, for the service in each case.
- (h) Rubber expansion joints shall be reinforced in each case to suit the particular location and service for which they are to be used. Rubber expansion joints shall not be used in piping where the working temperature exceeds 65°C.
- (i) A data sheet shall be submitted for each expansion, joint according to the Format in Exhibit 3.

4.5.2 Traps

- (a) All steam traps shall be of the latest improved design. All traps shall be made with cast steel or forged steel bodies, covers, etc.
- (b) All traps shall have metal working parts of materials suitable for the temperatures and pressures encountered. All seats and discs shall be removable.
- (c) All actuating levers and arms for the traps shall be of cast or forged steel. No pressed steel construction for these parts shall be used.
- (d) All traps shall be provided with properly valve bypasses and blowdown valves, unless specifically indicated otherwise.
- (e) A data sheet will be submitted for review prior to release for fabrication for each trap according to the format in Exhibit 7.

4.5.3 General Strainers

- (a) Strainers shall be provided where required by the application.
- (b) Strainers shall have the capability to withstand the design conditions or higher as required by the Bidder.
- (c) Strainer body construction shall comply with the requirement established in Section 4.5.5.
- (d) Strainer elements specification shall comply with the requirements established in Section 4.5.5 .
- (e) Data sheets shall be submitted for each strainer according to the format Exhibit 5. The data sheets will be reviewed by the Owner prior to release for fabrication.

4.5.4 Temporary Strainers

- (a) Temporary start-up strainer element (internal filter mesh) will be used at the suction of centrifugal pump where necessary.
- (b) Differential pressure gauges shall be furnished with each start-up strainer where necessary.
- (c) The temporary strainers shall be removed and replaced with the permanent in line strainers (as applicable) after commissioning at no additional cost to the Owner.

4.5.5 Automatic Backwashing Strainers

(a) Type and Design :

Each strainer shall be designed for continuous straining of foreign material from the water to be cleaned. Backwash cleaning of the strainer shall be automatic as specified below and shall backwash the strainer without interrupting its service.

The strainer should be capable of continuous service. Backwashing of the strainer shall be continuous or intermittent as required without any interruption of strainer operation. Strained water from the strainer outlet shall be used for backwash.

Any operation of the strainer shall, be motor-operated, including service flow control valves or backwash flow control valves.

(b) Construction :

Strainer bodies shall be cast iron or steel suitable for the design conditions.

Fabricated steel bodies shall have flanged or weld ends for pipe connections as required and subject to Owner review.

Strainer bodies shall be provided with suitable lifting attachments to enable the equipment to be lifted in its normal position. Lifting attachments shall be constructed to withstand any handling condition that might be encountered such as rapid lowering of sudden braking of load.

(c) Media :

Strainer elements shall have 18-8 stainless media sections of basket, cone or cylinder type. Media sections shall have openings as indicated in the data sheets approved by Owner. The elements shall be constructed to withstand maximum pressure differential.

Pressure drop shall be as indicated in the Data Sheet. Strainer size shall be selected to provide normal velocities of 1 .8 to 2.5 m/s resulting in an economic pressure drop.

(d) Control :

Bidder shall provide both automatic and local manual control of the strainer backwash cycle. Automatic backwash shall be initiated by strainer differential pressure, or by an adjustable 24-hour timer that will initiate backwash after a preset time period of strainer operation in the event that differential pressure backwash has not occurred. Local manual backwash shall also be provided and shall include a local

differential pressure gauge and a control station that shall initiate backwash located on the Bidder furnished control panel for each strainer. Initiation of a backwash by differential pressure shall reset the 24-hour timer. A lock-out feature to prevent backwash shall be provided in the event the strainer is removed from service.

Bidder shall provide alarm contacts with ratings as specified or required for alarming the following conditions on the Owner's enunciator :

- (i) Strainer differential pressure in excess of the automatic backwash set point.
- (ii) Strainer backwash cycle has failed to return to normal operating mode.

All differential pressure switches, backwash valves, solenoid valves, etc. required for automatic back washing of the strainer elements.

4.5.6 Flexible Hoses

Flexible hoses shall be furnished where required by Bidder Design.

The hoses shall be capable of withstanding the design conditions or higher if required by Bidder.

The hoses shall be of braided SS with flanged connections.

A data sheet shall be submitted for each flexible hose in accordance with the format Exhibit 2.

The data sheets shall be submitted for review to the Owner and a release obtained prior to procurement of the flexible hoses.

4.5.7 Restriction Orifice Plates

Restriction orifice plates shall be furnished where required by Bidders design.

The orifice plates shall in general be constructed of 304 SS and shall not be less than 6 mm thick.

A data sheet shall be submitted for each orifice plate in accordance with the requirements of Exhibit 4. The data sheets shall be reviewed by the Owner prior to release for fabrication.

4.5.8 Sight Flow Indicators

Sight flow indicators shall be provided as required for ease of operation.

A data sheet shall be submitted for each indicator per the format indicated in Exhibit 6.

The data sheets shall be reviewed by the Owner prior to their release for procurement.

4.5.9 Miscellaneous Piping

Miscellaneous piping specialties shall be provided as required by the Bidder's design.

These specialties shall have a minimum rating or higher as required by Bidder design.

A data sheet shall be submitted for each specialty per the format in Exhibit 1.

For the specialties shown on the P & ID's the data sheets shall be reviewed by the Owner prior to their release for fabrication.

4.6 Miscellaneous Accessories

Unless otherwise specified, design pressures and materials for accessories shall conform with design pressures and materials for valves in the lines in which the accessories are installed.

Vacuum seals shall be provided if required for the service.

A suitable compressed air filter-pressure regulator shall be provided at each piece of pilot-operated equipment.

4.7 Identification of Components

All piping components furnished separately by Bidder shall have a securely attached, metal tag marked to identify each item and its particular service. Tag materials shall be stainless steel. Identification shall include the individual pipe line, component numbers. Identification on drawings submitted for review, for specialties, shall include the same number used on the tag.

Tag attachments shall be as follows

- a. For traps and other specialties manufactured by others, excluding pipe and fittings: Permanently attached.
- b. For pipe, tubing, fittings and shop-fabricated piping sections: Temporarily attached.

All alloy steel pipe shipped in random lengths shall be color-coded with identifying color stripes running the full length of each pipe to prevent interchanging pipe grades during installation. In addition, each length and each piece cut from a length shall be marked with the applicable ASTM or ASME No., grade and heat number.

4.8 Fabrication

4.8.1 General Fabrication

Change of direction in welded pipeline shall be made by means of welding fittings (in lieu of pipe bends) except as follows:

Pipe bends shall be used where indicated on the design drawings.

Five diameter pipe bends may be used in pipelines 50 mm nominal size and smaller where practical and where clearances are adequate.

Pipe bends may be used only where accepted by the Owner.

Pipe bends, where used, shall conform to the requirements set forth in ANSI B31.1, in particular, Paragraph 102.4.5. The minimum wall thickness of pipe material for bends and the radius of bends shall be as set forth in Table 1 02.4.5 of ANSI B31 .1. The wall thickness given for pipe includes allowance for bends, unless otherwise noted therein. Pipe bends shall be made true to the specified radius, shall curve uniformly and shall be made in a manner that will preclude buckling of walls and/or flattening of the pipe cross section. Welds shall not be made in pipe bends unless reviewed and accepted by the Owner. If pipe lengths available necessitate a girth weld in a bend, the Owner shall be advised so that special weld end dimensions can be calculated.

Bidder shall be responsible for the proper design and fabrication of all piping system branch connections that are furnished as part of the Work. The rules governing the design of branch connections shall be in accordance with the applicable requirements of the governing code. All additional reinforcement required by the design conditions shall be provided by Bidder in the fabrication of the branch connections.

Welded branch connections and special welded fabricated fittings shall be carefully fitted and properly reinforced as required by the applicable codes.

Nozzles on headers to which stop valves, safety valves and relief valves are welded shall have inside and outside diameters which correspond to the inside and outside diameters of the welding ends of the valves to which they connect. Details of relief and safety valve nozzles shall be shown on the design drawings.

Threaded connection shall not be used unless called for in the design drawings.

All steel butt-welding fittings shall be as strong as the pipe to which they connect and shall be in accordance with the ANSI Standard for Steel Butt-Welding Fittings (ANSI B16.9).

Dimensions for short radius fittings, where such fittings are indicated on the design drawings, shall conform to applicable requirements of ANSI B16.28, except that the fittings shall be certified for a pressure rating fully equal to the design pressure of the pipe to which they connect.

Bidder shall consider carefully the type of welding fittings used where post heat treatment is required. In order that there be no interference with heat treating of field welds, the design shall provide sufficient room for the placement of local electrical stress relief apparatus or shall be provided with stub ends approximately 150 to 200 mm long.

Before fabrication the Bidder shall verify all measurements at Site and obtain all necessary additional information for the completion of the Work, including the following:

- a. Actual location of weld nozzles, flanges or other type terminal connections and verification of weld nozzle ends and flange facings in place, to which Bidder's work connects.
- b. exact location of existing piping with supports and hangers in place.
- c. Exact location of new and existing structures and equipment.
- d. Interferences or difficulties that may exist.
- e. Field measurements to allow for such make-up lengths or "closures" as may be necessary for accurate alignment and assembly. Details of such make-up lengths and/or closures required shall be submitted to the Owner for review prior to fabrication.

Terminals:

Where the Work connects to equipment or piping furnished in by Bidder, Bidder shall provide the proper terminal welding ends or flanges required on piping furnished by Bidder.

Insulation:

Insulation will be furnished per M-002 Technical Specification, Thermal Insulation & Lagging. Piping shall be arranged to allow clearance for insulation, as set forth in Part 4 Section 4, Inspection and Testing.

Weld metal shall be similar to and compatible with that of the piping and attachment.

Attachment welds shall be in accordance with the applicable requirements of the governing code, with procedures and welders qualified in accordance with ASME Section IX and acceptable to the Owner.

Guarding Devices at Pipe Sleeves and Pipe Openings.

Bidder shall provide all guarding devices for the pipe sleeves and pipe openings as necessary to meet national, OSHA and local requirements. All guarding devices shall be such as not to restrict any shifting of the pipelines during operating and shut-down conditions.

Flashing:

Bidder shall provide a flashing of 0.8 mm, Type 304 stainless steel around each pipe, the flashing extending well over the curb. Flashing shall be secured to pipe by means of a stainless steel clamping band with stainless steel bolts and nuts in such a manner as will make a rain tight joint.

Bidder shall also provide flashing for all pipelines passing through exterior walls, except where pipeline is anchored and/or sealed in the wall.

4.8.2 Shop Fabrication

All piping 80 mm in diameter and larger shall be fabricated and welded in the shop smaller piping shall be shop fabricated as far as possible, to reduce the number of field joints to a minimum.

The location of all shop and field welds in all major piping systems and in critical piping shall be shown on Bidder's shop drawings and shall be acceptable to the Owner.

The length of shop-welded sections shall be determined by shipping limitations, handle ability and erection space available, subject to review and acceptance by the Owner.

The ends of all weld connections on shop fabricated sections shall be properly machined.

Flanges, where required in high-pressure work, shall be attached in the shop. All flange attachments shall be rechecked after welding to ensure proper alignment.

Each shop-fabricated section of piping shall as far as possible have all nozzles, weld end fittings, and all coupling and nipples for drains -or- for instrument connections installed in the shop.

Piping shall be shop fabricated to suit the requirements of Part 4, Section 4, Erection, Commissioning, inspection and Testing.

4.8.3 Welding

4.8.3.1 General

All welding, inspection and testing of welds and welders shall be carried out in accordance with the applicable American, European, British and German Standard specifications (ASME, BS-EN, BS, DIN, AD-HP) as required by the design conditions of the plant and equipment and agreed by the Owner/Engineer.

Where there is conflict between Codes and/or Standards the Owner's decision will prevail.

The Bidder shall supply all necessary plant, labor, equipment, appliances, materials and consumables required to perform all operations in connection with the welding within the limits of this contracts. All equipment, plant and appliances shall be good quality, well maintained and entirely suitable for the intended purpose.

4.8.3.2 General fabrication

The Owner or his Representative shall be notified at least two days prior to commencement of any assembly or fabrication work on site.

The CO₂ and flux-cored welding process will not be acceptable for site welding unless otherwise approved by the engineer.

Welders and weldings shall be protected by suitable equipment from wind, airborne sand and moisture. Work shall not be performed when weather conditions do not permit satisfactory workmanship or adequate inspection.

Weld preparation profiles may be prepared by machining, shearing, flame cutting and grinding and shall be performed carefully and accurately. Flame cutting is prohibited for stainless steel and high alloyed steel (>5% alloy quality). Whenever possible a mechanically guided tool shall be used for flame cutting.

Before execution of the welds, the profile shall be completely clean that means dry, free of any kind of humidity, slag, oxides, paint etc.

Where preheating is required it shall whenever practical be carried out by electric resistance or induction equipment. Preheat and interpass temperatures shall be checked by temperature indicating crayons at a distance of not less than 75 mm from the weld groove. The use of hand-held torches for preheating will not be permitted.

Post weld heat treatment in accordance with the applicable European, British and American Standards shall be carried out using equipment specifically designed for the process. Gas burners or hand held torches will not be permitted. For work which is heat treated in a furnace, the Bidder shall supply copies of the heat treatment charts. Work which is subjected to heat treatments by means of electrically heated elements shall be monitored by suitable thermocouples (minimum two thermocouples) connected to a chart recorder.

Copies of heat treatment charts shall be made available to the Owner/Engineer.

While post weld heat treatment, the pipes shall be extremely supported to ensure that no stress is coming on the weld. All post weld heat treatments shall be carried out under the supervision of qualified personnel.

All transition welds between dissimilar materials shall be made under controlled conditions at the manufacturer's work. Site welding of structural high yield steel will not be permitted without the prior consent of the Owner or his Representative.

All tube butt welds shall have 100% penetration. The surface of the finished weld shall not be concave.

Stainless steel shall be welded under a local inert atmosphere, including gas backing.

All stainless steel welds shall be cleaned and passivated.

4.8.3.3 Weld procedure documents

Complete and fully detailed weld procedure documents shall be kept and these shall be made available to the Engineer on request.

Prior to the commencement of welding the Bidder shall submit to the Engineer for approval the welding procedures to be used in the fabrication of the relevant sections of work.

The weld procedure documents shall be fully detailed and each shall indicate clearly which item it is intended to cover. The procedures shall be in accordance with the requirements of BS 499 Part 1, Appendices A-G.

Weld procedure sheets for different materials of piping and valves shall be specified in appropriate sections of Instruction Manuals.

4.8.3.4 Weld procedure qualification tests

Weld procedure qualification tests shall be carried out in accordance with the requirements of Section IX of the ASME Boiler and Pressure Vessel Code or British Standard 4870 / BS EN 288 or agreed National Standard for the item of plant under consideration. Providing that the Bidder confirms that the basic parameters of the procedure have not been changed since approval, the results of weld procedure qualification tests previously carried out under the Supervision of any internationally recognized inspecting authority may be accepted by the Owner or his Representative.

The Bidder shall inform the Owner or his Representative of any proposed changes to the welding procedures before such changes are implemented. If in the opinion of the Owner or his Representative a further qualification test is required as a result of such changes, then the Bidder shall perform the required test without additional charge.

The results of all tests shall be made available, for examination by the Owner or his Representative, if required.

4.8.3.5 Welders qualification tests

All welder's and welding operators shall be qualified for the work and shall hold current welders qualification certificates in accordance with British Standard 4871 / BS EN 729 or agreed National Standard for the work.

All welders tests shall be witnessed and approved by the Owner or his Representative before the welder or operator is permitted to work. The decision of the Owner or his Representative regarding the acceptability of any test or existing qualification tests shall be accepted solely at the discretion of the Owner or his Representative.

A welder or welding operator who fails one or more of the tests shall be retested after further training. Before retest, Bidder should confirm that the failed welder or welding operator has had further training and practice.

Records showing the date and results of the qualification tests performed by each welder and weld operator together with the identification number assigned to him shall at all times be available for scrutiny by the Owner or his Representative.

The detailed welders summary list shall be submitted to the Owner or his Representative at agreed intervals of time regularly.

4.8.3.6 Storage of welding consumables

Welding consumables shall be stored in a manner that will protect them from all forms of deterioration prior to use and shall be properly identified.

Each batch of electrodes and wires shall be identifiable with the respective manufacturer's test certificate and advice note and shall be used in strict delivery rotation.

Electrodes which have damaged flux coatings are bent, misshapen or have lost their identification shall not be issued on any work.

The Bidder shall provide drying ovens in sufficient number to permit the correct storage of electrodes for 48 hours before use. Drying ovens shall preferably be heated by electric means and shall have automatic heat controls and visual temperature indication.

The re-baking of low hydrogen electrodes shall be done strictly in accordance with the manufacturer's recommendations.

The procedure for the issue of electrodes shall be carefully controlled. No welder shall be issued with more electrodes than can be used during a ½ day work period.

As far as necessary, all electrodes shall be issued and transported to the work place in thermally insulated containers. The electrodes shall remain in the container until they are required.

Gas cylinders for use with burning or welding equipment shall be marked in accordance with the requirements of BS 349 (ISO/R448).

4.8.3.7 Welding equipment

Any welding equipment which, in the opinion of the Owner or his Representative, is unsuitable or unsatisfactory for the service for which it is being used, shall be replaced by the Bidder.

The absence of comment by the Owner or his Representative shall not be taken as approval for any equipment.

4.8.3.8 Weld examination

All non-destructive examinations shall be supervised by a fully qualified and experienced specialist appointed by the Bidder. Individual operators in each of the respective techniques shall be qualified and trained in the respective subject and shall have reached a standard comparable with the Certification Scheme of Welding Inspection Personnel.

Testing shall be in accordance with the requirements of British Standard 709 "Methods of Testing Fusion Welded Joints and Weld Metal in Steel" the European equivalent or an agreed National Standard.

4.8.3.9 Visual weld inspection

Each weld shall be subjected to a stringent visual inspection and shall be free from undercut, excessive spatter, craters, cracks, porosity and other surface imperfections. Welds shall be of regular contour, even weld ripple and indicative of good workmanship.

Fillet welds shall be checked for dimensional tolerance and form using a fillet weld gauge. Fillet welds should be slightly concave in form and each leg of the weld shall have equal length.

4.8.3.10 Internal examination

The internal root bead of tube butt welds shall be examined by intrascope or other suitable optical device.

4.8.3.11 Radiography

(a) General

Radiography shall be carried out in accordance with the requirements and recommendations of the appropriate Standard as agreed by the Owner or his Representative.

(b) Manufacturer's works

Radiographic inspection at manufacturer's work shall be accomplished with X-ray equipment throughout.

(c) Site

Gamma radiography will be permitted in appropriate cases at the job site. Cobalt 60 sources shall not be used without the express permission of the Owner or his Representative in writing.

(d) Techniques

Radiographic techniques shall be in accordance with ASME & British Standard BS EN 1435 "Non-Destructive Examination of Welds - Radiographic Examination of Welded Joints" and other relevant International Standards or agreed National Standards.

The Bidder shall take all possible steps to protect radiographic and all other personnel from the effects of ionizing radiation.

4.8.3.12 Ultrasonic examination

Ultrasonic examination of welds shall be carried out in accordance with British Standard 3923, Part 2 "Automatic Examination of Fusion Welded Butt

Joints in Ferric Steels"; BS EN 1714" Non-Destructive Examination of Welds - Ultrasonic Examination of Welded Joints"; and any other relevant British Standards or agreed National Standards.

4.8.3.13 Magnetic crack detection

Magnetic crack detection shall be carried out in accordance with BS 6072 "Methods for Magnetic Particle Testing of Welds". or an agreed National Standard.

4.8.3.14 Dye penetrant tests

Dye penetrant tests shall be in accordance with British Standard BS 6443 "Method for Penetrant Flow Detection" or any other relevant British or agreed National Standards.

4.8.3.15 Quality requirements for welds

All welds subjected to non-destructive tests shall be entirely free from cracks or crack like defects, lack of root fusion, lack of sidewall fusion, root bum through, or tailed pores. The standard for porosity and slag inclusions will be as indicated in the agreed standards for design and welding.

Personnel who perform non-destructive examination of welds shall have qualified and certified in each examination method in accordance to that standard, which is used for the welding and examination processes.

4.8.3.16 Mandatory inspections

A minimum of 10% of all butt welds on all classifications of work shall be radiographically examined, unless otherwise agreed with the Engineer.

In addition to the minimum of 10% examination, all transmission welds between dissimilar materials, such as high alloy steels to carbon steel, or austenitic steels or non ferrous materials to steels, shall be subjected to 100% ultrasonic examination or crack detection wherever practicable. In addition, all butt welds between dissimilar materials shall be subjected to 100% radiographic examination.

Also, all welds in ferritic alloy steels, e.g. having a carbon equivalent value in excess of 0.40% and high yield-strength steels, e.g. having a yield strength greater than 300 MPa, shall be subjected to 100% ultrasonic examination and crack detection wherever possible. In addition, all butt welds in these materials shall be subjected to 100% radiographic examination.

A minimum of 10% of all fillet welds on all classifications of work shall be examined by magnetic crack or dye penetrant tests, unless otherwise agreed with the Engineer.

The minimum weld examination shall be spreaded to all welders, that approximately 10% of its welds are tested.

4.8.3.17 Weld repairs

The Engineer's approval shall be obtained prior to commencement of any repair or rectification work.

Weld repairs shall be made to the same procedure as for the original weld. All tests shall be repeated after the repair has been completed and reports on radiographic and ultrasonic tests shall be marked to indicate that the report refers to a repaired weld.

4.8.4 Erection of Piping

Erection of piping shall be in accordance with the applicable requirement of this Specification.

4.8.5 Flanged Joints

4.8.5.1 Contact Face Finish

Contact faces shall be set forth by the Bidder for each particular service. Contact faces for all male and female, and tongue and groove faced flanges shall have smooth tool finish, unless otherwise indicated.

4.8.5.2 Gaskets

Gaskets for all joints shall be of such inside dimensions that no portions will project into the ports of valves, pipes or fittings. Gaskets for male and female joints, and for tongue and groove joints shall fit into the female or groove facing with approximately clearance.

Gaskets for joints shall extend to within of the bolts. Plain faced flanged joints shall have full face gaskets with holes for bolts, unless otherwise indicated.

Gaskets shall be set forth by the Bidder for each particular service.

When ordering spiral-wound metal gaskets, Bidder shall specify complete joint data for each specific application, including:

1. Service and operating pressures and temperatures.
2. Flange material and pressure rating.
3. Type of flange facings.
4. Bolting materials.

4.8.5.3 Studs

All alloy steel studs shall have identification marks at one end, covering material, class etc. so that studs for low temperature and high temperature work will not be interchanged during installation.

All steel bolts shall have Class 2A fit threads and all nuts for steel bolts shall have Class 2B fit threads, per ANSI B 1.1.

Standard metric nuts and bolts are also acceptable per standards reviewed by - the Owner.

4.8.5.4 Thread Compound

Threads of all bolts and studs shall be painted with a suitable thread compound before joint is made up. The compound shall be suitable for the operating temperatures involved. For high temperature service special high temperature thread compound must be used.

4.8.5.5 Installation of Studs for High-Pressure Flanged Joints

Unless otherwise specified, all studs for flanged joints in high-pressure steam piping and high-pressure boiler feed piping in sizes 80 mm and larger shall be micrometered when installing in order to determine the elongation and initial stress in the studs.

Studs for the above piping shall be installed with an initial compression suitable to force the gasket material into intimate contact with the joint faces. The initial stud bolt tension required to set the gasket in place shall be in accordance with the best current practice and as dictated by the flange facing and gasket material involved: Initial bolt tension to secure the desired bolt stress shall be determined by means of measuring the elastic elongation of the studs. If heating of studs is used to obtain the initial tension, the method of heating and the maximum temperature obtained on the studs shall be subject to the Owner review and acceptance.

Bidder shall check the bolt tension in each flanged joint made up by him, within two weeks after each line in the above piping is up to the full temperature. In doing this, one stud for each four studs in the joint shall be released to determine the residual elongation, but not more than four studs in any one joint shall be released to obtain the residual elongation. It shall be assumed that all other studs in the same joint have a residual elongation equal to the average obtained from the released studs. After the residual elongation of the studs in the joint has been determined, each stud shall be pulled up to the elongation required to produce the initial tension specified above. Not more than one stud in a joint shall be released at one time and this stud shall be retightened to the initial tension before another stud is released.

The bonnet joints of all valves installed by Bidder in the above mentioned high-pressure steam and high-pressure boiler feed piping shall be “followed up” in the same manner as is described above for the line joint flanges.

4.8.6 Pipe Supports and Hangers

Pipe Supports and Hangers shall be furnished in accordance with M-002 Pipe Supports and Hangers Technical Specification.

4.9 Piping Line Number

Piping lines larger than or equal to 80 mm in diameter shown on the Piping and Instrument Diagrams (P & ID) and the physical drawings shall have discrete line numbers. For systems with no line sizes greater than or equal to 80 mm a typical line will be chosen and all information noted in the line list. These numbers, in each case, shall be encircled on all drawings. The line number shall indicate the plant unit number(s), the system identification, the line serial number and the pipe size. A change in pipe size, pressure class, materials or code jurisdiction etc. is denoted by a change in line number. The line numbers are to be used for identification of lines in all drawings,

documentation and correspondence. The line numbers shall be tabulated in a piping line list to be prepared by the Bidder in accordance with this Specification. For further details about numbering see Part 4, Subsection 2.8, Plant Identification.

4.10 Piping Line list

Each pipeline number shall appear on the piping “Line List”. The Line List shall designate information required for design, fabrication and erection of all piping.

The line list shall have a minimum of fifteen (15) field of information, as indicated below:

Field 1	-	Number
Field 2	-	Line Number
Field 3	-	Line Size
Field 4	-	Description
Field 5	-	P & I Diagram No.
Field 6	-	Max. Operating Pressure bar (g) Temperature ° C
Field 7	-	Design Pressure bar (g) Temperature ° C
Field 8	-	Piping Class
Field 9	-	Piping Design Table
Field 10	-	Instrument Class
Field 11	-	Instrument Design Table
Field 12	-	Insulation Type Thickness
Field 13	-	Field T.P.
Field 14	-	Remarks
Field 15	-	Revision

4.11 Cleaning, Protection

4.11.1 Cleaning

(a) General

It is highly important that the inside of all piping be free of loose scale, welding icicles and spatter, oil, grease, rust, paint, welding fluxes, chalk, abrasives, carbon deposits, coatings, moisture and all other foreign

matter before it is placed in service. Cleaning shall be done in accordance with the following :

(b) Fabricated Principal Piping

After fabrication, the interior surfaces of all critical piping systems and those principal piping systems, except lube oil systems, shall be thoroughly cleaned by one of the following methods.

(c) Ferritic Steel Piping

Acid picking in a suitable solution for a sufficient time to loosen all scale. After picking and thorough drying the pipe shall be metal shot-blasted, grit-blasted or turbinized to remove all scale, after which it shall be neutralized and rinsed out with hot water at about 82°C. After rinsing, the pipe shall be blown out with compressed air. The air used for final blowing out must be free from all moisture.

Blasting to a white, metal finish with angular steel grit of suitable size after - which piping shall be blown out with moisture free compressed air. In order to avoid the possibility of thinning the pipe wall when blasting, particular care must be taken so that the jet will not be allowed to impinge on any portion of pipe for a longer period of time than is required to thoroughly clean the interior of pipe. The gritblasting and post-cleaning shall be performed in accordance with the applicable procedures for No. 5 White Metal Blast Cleaning as set forth in Steel Structures Painting Council, SSPC-SP5-63. Photographic standards of comparison may be used to define the final surface condition to be supplied under SSPC-SP5-63. For intact mill scale, for partially rusted and pitted surfaces, the appearance of the surface after White Metal Blast Cleaning shall correspond with pictorial standards ASa3, BSa3, SCa3, DSa3 of SSpc - Vis 1-63T.

(d) Austenitic Stainless Steel Piping

Pickling as specified above except pickling and neutralizing solutions and procedures shall be specially suitable for austenitic stainless steel and shall be acceptable to the Owner.

(e) Fabricated Lube Oil Piping

After fabrication, acid pickle the inside surfaces with a suitable solution for a sufficient time to loosen all scale, rust, etc.

After pickling and thoroughly drying, rinse with clean water at about 82°C until all contamination is removed and treat with suitable neutralizing and surface conditioning agent. After rinsing, blow out with compressed air. The air for final blowing must be free from moisture.

(f) Bends after Fabrication

The interior of all bends, except those for lube oil systems, shall also be thoroughly cleaned by the methods specified above.

(g) Other Piping

The interior of all other pipe, fittings, valves, specialties, fabricated piping and appurtenances shall also be thoroughly cleaned by suitable method acceptable to the Owner. As a minimum, the materials shall be rapped and blown out with compressed air.

(h) Piping Materials Furnished by Bidder :

Fabrication by Bidder :

Where pipe, fittings, flanges etc. are furnished by Bidder for fabrication by Bidder, Bidder shall, after fabrication, clean such piping in accordance with the requirements herein before specified.

Erection by Bidder :

Where valves and other specialties and fabricated piping are furnished by others for erection by Bidder, such materials will be shop cleaned by other; however, Bidder shall inspect the inside of such materials, prior to installation, to ensure that they are clean. Bidder shall reclean such materials, if necessary.

Reused Materials :

All pipe, fittings, valves and other appurtenances (if any) which have been removed by Bidder from present systems and which are to be reused by Bidder, shall be thoroughly cleaned by him both inside and outside and all working parts of valves and other operating appurtenances shall be checked by Bidder for proper functioning before they are reinstalled. Any malfunctions shall be reported to Owner's representative.

(i) Cleaning During and After Erection

Bidder shall ensure that all piping erected by him is maintained clean during erection.

Bidder shall clean out the inside of all cut-ins and openings made by him in existing piping for new connections. Bidder shall blow out existing piping, prior to making new connections, if required, as directed by Owner's representative.

Bidder shall perform blowing out, flushing, chemical cleaning and other special cleaning operations on principal systems after erection, where called for in the Specification, as directed by Owner's representative and/or as indicated on the drawings.

(j) Temporary Piping and Materials

Bidder shall furnish lubricating and flushing oils, chemical solutions and steam and water source for special field cleaning after erection.

Compressed air, where required for field blowing out, shall be furnished by Bidder.

Bidder shall furnish and install all other temporary materials, equipment and facilities required for cleaning, blowing out, flushing etc. to be done by Bidder, including valves, other piping, pumps, compressors, heaters, instruments and controls.

(k) Closures

All openings shall be provided with temporary closures, as hereinafter specified, as soon as possible after cleaning.

(l) Cleaning Materials and Procedures

Prior start of the Work, Bidder shall submit to the Owner for review, his methods and procedures for cleaning the piping and type of acid and neutralizing agent to be used for pickling.

4.11.2 Protection

Materials shall be provided with protection against damage, corrosion and internal contamination in accordance with the following:

4.11.2.1 Materials shipped to the Site

All materials and equipment shall be packaged, packed or prepared for shipment in a manner which will ensure arrival at destination in satisfactory condition. Procedure and details shall be submitted to the Owner for review prior to start of shipment.

All openings in piping and equipment shall be securely plugged, capped or otherwise blanked off, sealed with tape and suitably protected against damage and entry of foreign materials and moisture. For shop fabricated piping, this shall be done as soon as possible after shop cleaning.

Weld ends on all valves, fittings, pipes, nozzles, etc., shall be capped and sealed with suitable, firmly attached protectors.

Butt-weld ends on ferrous materials shall be coated with deoxialuminate, or acceptable equal, prior to capping, back to a ring 150 mm from the end.

Protectors for beveled ends designed for backing ring or consumable insert shall have a plywood or hard board liner disc held securely against the beveled end. Protectors for other weld ends may be metal caps without liner discs.

All flange facings, bolt holes and other machined surfaces of ferrous materials (except butt-weld end preparation) shall be coated with a suitable removable antitrust compound. No coatings shall be applied to nonferrous materials.

All flanged connections and loose flanges shall be provided with suitable full face flange protectors bolted in place and sealed.

All protectors for openings and all braces, brackets, spacers, ties, bindings and other shipping, packaging and packing materials and appurtenances used for protection in shipping, storing and handling of nonferrous piping and materials shall be of such design, type and/or arrangement as to prohibit contact between ferrous and nonferrous materials.

Each section of pipe for lube oil system shall contain a bag of desiccant fastened to the center of masonite liner provided on one of the opening protectors; and all opening protectors shall be sealed with a vapor barrier and taped, and the tape varnished.

4.11.2.2 Materials at the Project Site

Materials shall be handled, stored and protected in such a manner as will prevent damage to it and prevent entry of foreign material and moisture.

Materials shall be stored above grade.

Bidder shall ensure that all piping openings are provided with temporary protectors securely attached and sealed, up to the time of field fabrication or erection.

For piping field fabricated by Bidder, all openings of the fabricated piping and remaining cut pipe shall be provided with protectors, as soon as possible after cleaning. Protectors shall be similar to those specified above and shall remain in place until removal is necessary for connection thereto.

Bidder shall ensure that the associated terminal connections are provided with protectors. Where such terminals are located in an area where they may be subject to damage from other operations at the Project Site or constitute a hazard, Bidder shall provide barricades and/or warning signs around the terminals.

Bidder shall, at the end of each working day, ensure that all remaining openings in piping fabricated by him and all new openings made by him in existing piping are provided with temporary protectors.

5 PERFORMANCE TESTING AND GUARANTEES

Performance Testing and Guarantees are included in Clause 6, Quality Assurance/Quality Control of this Specification.

6 QUALITY ASSURANCE/QUALITY CONTROL (Q.A/QC)

Quality Control shall conform to the requirements of QA-MTCCPP included in Part 4, Subsection 2.7.

The quality control procedures and the quality control documentation as specified in shall be provided by the Bidder.

EXHIBIT 1 UNIVERSAL DATA SHEET FOR MISCELLANEOUS PIPING SPECIALTIES

PIPING SPECIALTY NUMBER													
SERVICE													
1.													
2.													
3.													
4.													
5.													
6.													
7.													
8.													
9.													
10.													
11.													
12.													
13.													
14.													
15.													
16.													
17.													
PLANT DATA	SAF. CL/ SEIS.CAT												
	QA/QUAL. GR												
	ELECT. CLASS												
FURN. SPEC./INSTALL.SPEC												P&ID	
RELEASE RECORD										MUARA TAWAR CCPP			
REV	DATE RELEASED	PREPARED	REVIEWED	APPROVED	REV	DATE RELEASED	PREPARED	REVIEWED	APPROVED	UNIVERSAL DATA SHEET FOR MISCELLANEOUS PIPING SPECIALTIES		DATA SHEET	

EXHIBIT 2 FLEXIBLE HOSES

NUMBER		1	2	3	4						
SERVICE											
1.	HOSE DIAMETER (ID)										
2.	HOSE DIAMETER (OD)										
3.	FLUID										
4.	PRESSURE, DESIGN/NORMAL										
5.	TEMPERATURE , DESIGN/NORMAL										
6.	FLOW RATE , DESIGN/NORMAL										
7.	LENGTH REQUIRED										
8.	BRAIDING										
9.	MINIMUM BENDING RADIUS										
10.	END CONNECTION OR FITTING										
11.	MATERIAL OF CONSTRUCTION										
12.	MANUFACTURING										
13.	MODE OR TYPE										
14.	SPECIAL REQUIREMENTS										
PLANT DATA	SAF. CL/ SEIS.CAT										
	QA/QUAL. GR										
	ELECT. CLASS										
FURN. SPEC./INSTALL.SPEC											
RELEASE RECORD										MUARA TAWAR CCPP	
REV	DATE RELEASED	PREPARED	REVIEWED	APPROVED	REV	DATE RELEASED	PREPARED	REVIEWED	APPROVED	FLEXIBLE HOSES	
										DATA SHEET	

EXHIBIT 3 EXPANSION JOINTS

NUMBER		1		2		3		4			
SERVICE											
1.	PIPE DIA/SCHEDULE										
2.	FLUID										
3.	PRESSURE, DESIGN/NORMAL										
4.	TEMPERATURE , DESIGN/NORMAL										
5.	FLOW RATE , DESIGN/NORMAL										
6.	MAXIMUM LATERAL MOVEMENT										
7.	MAXIMUM AXIAL MOVEMENT										
8.	POSITION OF EXPANSION JOINT										
9.	TIED OR UNITED										
10.	MATERIAL OF CONTRUCTION										
11.	MANUFACTURING										
12.	MODEL NUMBER										
13.	SPECIAL REQUIREMENTS										
PLANT DATA		SAF. CL/ SEIS.CAT									
		QA/QUAL. GR									
		ELECT. CLASS									
FURN. SPEC./INSTALL.SPEC											
RELEASE RECORD										MUARA TAWAR CCPP	
REV	DATE RELEASED	PREPARED	REVIEWED	APPROVED	REV	DATE RELEASED	PREPARED	REVIEWED	APPROVED	EXPANSION JOINTS	
										DATA SHEET	

EXHIBIT 4 RESTRICTION ORIFICES

NUMBER		1	2	3	4						
SERVICE											
1.	PIPE DIA/SCHEDULE										
2.	FLUID										
3.	PRESSURE, DESIGN/NORMAL										
4.	TEMPERATURE , DESIGN/NORMAL										
5.	FLOW RATE , DESIGN/NORMAL										
6.	DIFF. PRESSURE , DESIGN/NORMAL										
7.	SPECIFIC GRAVITY										
8.	STEAM QUALITY										
9.	PLATE TYPE										
10.	PLATE MATERIAL										
11.	SIZE OF ORIFICE										
12.	TYPE & RATING OF FLANGES										
13.	SPECIAL REQUIREMENTS										
PLANT DATA	SAF. CL/ SEIS.CAT										
	QA/QUAL. GR										
	ELECT. CLASS										
FURN. SPEC./INSTALL.SPEC											
RELEASE RECORD										MUARA TAWAR CCPP	
REV	DATE RELEASED	PREPARED	REVIEWED	APPROVED	REV	DATE RELEASED	PREPARED	REVIEWED	APPROVED	RESTRICTION ORIFICES	DATA SHEET

EXHIBIT 5 STRAINER

PIPING SPECIALTY NUMBER											
SERVICE											
1.	PIPE DIA/SCHEDULE										
2.	STRAINER TYPE										
3.	FLUID										
4.	PRESSURE, DESIGN/NORMAL										
5.	TEMPERATURE , DESIGN/NORMAL										
6.	FLOW RATE , DESIGN/NORMAL										
7.	ALLOWABLE P DESIGN FLOW										
8.	P MEASUREMENT GAUGE										
9.	INLET LOCATION										
10.	OUTLET LOCATION										
11.	REQUIRED MESH SIZE										
12.	BODY MATERIAL										
13.	BASKET STRAINER MATERIAL										
14.	BACKWASH/DRAIN BLOCK VALVE										
15.	MANUFACTURER										
16.	MODEL NUMBER										
PLANT DATA		SAF. CL/ SEIS.CAT									
		QA/QUAL. GR									
		ELECT. CLASS									
FURN. SPEC./INSTALL.SPEC											P&ID
RELEASE RECORD										MUARA TAWAR CCPP	
REV	DATE RELEASED	PREPARED	REVIEWED	APPROVED	REV	DATE RELEASED	PREPARED	REVIEWED	APPROVED	STRAINER	
										DATA SHEET	

EXHIBIT 6 SIGHT FLOW INDICATORS

PIPING SPECIALTY NUMBER											
SERVICE											
1	PIPE DIA/SCHEDULE										
2	STRAINER TYPE										
3	FLUID										
4	PRESSURE, DESIGN/NORMAL										
5	TEMPERATURE , DESIGN/NORMAL										
6	FLOW RATE, DESIGN/NORMAL										
7	FLOW RATE, DESIGN/NORMAL										
8	FLUID CLARITY										
9	END CONNECTION										
10	BODY MATERIAL										
11	FIELD MOUNTING POSITION										
12	MANUFACTURING										
13	MODEL NUMBER										
14	SPECIAL REQUIREMENTS										
PLANT DATA	SAF. CL/ SEIS.CAT										
	QA/QUAL. GR										
	ELECT. CLASS										
FURN. SPEC./INSTALL.SPEC											P&ID
RELEASE RECORD										MUARA TAWAR CCPP	
REV	DATE RELEASED	PREPARED	REVIEWED	APPROVED	REV	DATE RELEASED	PREPARED	REVIEWED	APPROVED	SIGHT FLOW INDICATORS	
										DATA SHEET	

EXHIBIT 7 DATA SHEET FOR STEAM TRAPS

PIPING SPECIALTY NUMBER											
SERVICE											
	MANUFACTURER										
	MODEL NO										
	TRAP TYPE										
	CONDENSATE FLOW										
	INLET PRESSURE										
	BACK PRESSURE										
	NOMINAL SIZE										
	DESIGN TEMPERATURE										
	DESIGN PRESSURE										
	TYPE OF END CONN										
	MATERIAL OF CONST:										
	1. BODY										
	2. MAJOR INTERNALS										
PLANT DATA	SAF. CL/ SEIS.CAT										
	QA/QUAL. GR										
	ELECT. CLASS										
FURN. SPEC./INSTALL.SPEC											
RELEASE RECORD										MUARA TAWAR CCPP	
REV	DATE RELEASED	PREPARED	REVIEWED	APPROVED	REV	DATE RELEASED	PREPARED	REVIEWED	APPROVED	DATA SHEET FOR STEAM TRAPS	
										DATA SHEET	

(Last page of M-001)